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Synopsis On

Awen AI IOT ESP32 based health monitoring system with physiological and behavioural patterns Computer Science and Engineering (Data Science)

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ABSTRACT

AWEN (Adaptive Wellness & Emotional Navigation) is an integrated embedded monitoring and data-handling system developed around the ESP32 microcontroller. Monitoring physical parameters effectively in academic and experimental environments often presents a challenge when sensor data is confined to raw serial logs, fragmented displays, or unnecessarily complex, high-latency remote server platforms. AWEN addresses this challenge by providing an end-to-end, self-contained architecture that combines multi-sensor acquisition, immediate on-device feedback, a robust serial communication pipeline, an organized web-based monitoring dashboard, and structured local database persistence using SQLite.

The physical hardware layer interfaces three confirmed sensing components: the MAX30100 optical pulse and oxygen-related sensor for pulse and oxygen saturation estimation, the MPU6050 six-axis inertial motion sensor (three-axis accelerometer and three-axis gyroscope) for kinematic movement observation, and the LM35 precision linear analog temperature sensor for thermal monitoring. The central ESP32 microcontroller executes custom C++ firmware developed within the Arduino IDE. The firmware executes non-blocking acquisition routines, applies unit conversions, drives a local 0.96-inch I2C OLED display for immediate real-time visual inspection, and activates an audible piezo buzzer when monitored parameters exceed preset safety thresholds.

For structured record keeping and retrospective analysis, the ESP32 streams serialized telemetry packets over a USB-UART connection to a host computer. A web-based dashboard visualizes the incoming data through an intuitive graphical user interface, while a local SQLite relational database archives timestamped readings for persistent query and historical review. The entire system operates strictly as a self-contained local embedded unit without external internet dependencies or clinical diagnostic assertions. This synopsis outlines the technical background, literature survey, proposed methodology, hardware-software workflows, project scheduling, resource feasibility, and expected engineering deliverables of the AWEN prototype.

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Abbreviations and Symbols

Abbreviation	Full Description & Technical Role in AWEN Hardware
ESP32	Espressif 32-bit Dual-Core Microcontroller with Integrated Peripherals
I2C	Inter-Integrated Circuit Two-Wire Serial Communication Bus Protocol
ADC	Analog-to-Digital Converter (12-bit Successive Approximation Register on ESP32)
GPIO	General-Purpose Input/Output Pins for Microcontroller Interfacing
OLED	Organic Light-Emitting Diode (0.96-inch Monochrome Display, SSD1306)
UART	Universal Asynchronous Receiver-Transmitter (Serial Hardware Interface)
USB	Universal Serial Bus (5V Power Delivery and Data Link)
MEMS	Micro-Electro-Mechanical Systems (Miniaturized Kinematic Transducers)
PPG	Photoplethysmography (Optical Volumetric Measurement of Arterial Blood)
SpO2	Peripheral Capillary Oxygen Saturation Percentage
BPM	Beats Per Minute (Heart / Pulse Rate Frequency Unit)
MPU	Motion Processing Unit (6-Axis Tri-Axial Accelerometer and Gyroscope)
LDO	Low-Dropout Voltage Regulator (Step-Down 5.0V to 3.3V DC Rail)
SMD	Surface Mount Device (Packaging for Integrated Circuit Sensors)
TO-92	Transistor Outline 92 Plastic Encapsulation (Precision Temperature Sensor)
DFD	Data Flow Diagram (Structural Hardware and Signal Pipeline Modeling)
WBS	Work Breakdown Structure (Hierarchical Project Task Organization)
BOM	Bill of Materials (Hardware Component Specifications and Quantities)
SDG	Sustainable Development Goals (United Nations 2030 Agenda)
SJCEM	St. John College of Engineering and Management, Palghar

1. Introduction

1.1 Background

The continuous observation of human physical state and environmental interactions represents one of the most critical domains in modern instrumentation engineering. With the rapid evolution of microelectronics, low-power embedded computing, and advanced semiconductor sensors, it has become technically feasible to construct compact benchtop monitoring prototypes capable of continuous multi-channel physical signal acquisition. Traditionally, physiological and kinematic tracking systems were confined to specialized laboratory benches or high-cost clinical workstations. These legacy systems typically required bulky tethered power supplies, expensive analog signal conditioning amplifiers, and proprietary instrumentation suites that made decentralized or educational deployment impractical.

In recent years, modern 32-bit microcontrollers, particularly the Espressif ESP32 system-on-chip, have fundamentally reshaped the field of embedded hardware prototyping. Combining high-speed dual-core processing (operating at up to 240 MHz), rich hardware peripheral interfaces (including multiple hardware I2C buses, high-resolution successive approximation analog-to-digital converters, and hardware UART controllers), and low quiescent current draw, the ESP32 provides a robust hardware foundation for executing concurrent data acquisition and peripheral driving. This hardware capability allows engineers to interface heterogeneous sensor modalities—ranging from digital optical sensors and micro-electro-mechanical systems (MEMS) inertial measurement units to precision analog thermal sensors—within a unified, electrically coordinated benchtop architecture.

The AWEN (Adaptive Wellness & Emotional Navigation) hardware project is conceived as an integrated, self-contained multi-sensor monitoring apparatus centered on the ESP32 microcontroller. Rather than treating physical sensors in isolation or relying on fragile, high-latency wireless telemetry, the AWEN hardware prototype establishes a reliable physical computing ecosystem. The physical system acquires three fundamental physical parameters: optical pulse and oxygen-related indicators via the MAX30100 optical sensor, tri-axial kinematic acceleration and angular rotation via the MPU6050 6-axis motion sensor, and surface/ambient thermal levels via the LM35 precision linear analog temperature sensor. By combining immediate on-device visual feedback on a 0.96-inch OLED display, instant audible alert notification through a piezo buzzer, and structured telemetry transmission over USB-UART, the AWEN hardware prototype demonstrates a robust, multidisciplinary engineering realization.

1.2 Problem Statement

Despite significant advancements in sensor manufacturing and microcontroller availability, student and research prototypes in multi-sensor monitoring continue to suffer from electrical noise vulnerability, bus contention, and lack of immediate on-device user feedback. The problem addressed by the AWEN hardware prototype is structured across four key dimensions:

- **Existing Problem in Prototype Design:** In most academic and hobbyist embedded projects, sensor readings are routed exclusively to a PC screen via raw serial prints, leaving the physical unit devoid of local visual displays or immediate acoustic alarms. When disconnected from a host workstation, the physical device becomes completely inoperable.
- **Industrial & Technological Issue:** Commercial physiological monitoring devices increasingly lock physical telemetry behind proprietary closed hardware ecosystems with non-serviceable

components and non-standard pinouts. This prevents educational institutions and engineering laboratories from inspecting raw sensor signals, verifying calibration curves, or adapting hardware for custom laboratory experiments.

- **Identified Research Gap:** A noticeable gap exists in the literature between simplistic single-sensor embedded sketches that suffer from bus lockups and bulky medical workstations that are economically inaccessible. There is an acute lack of self-contained, dual-output benchtop prototypes that combine digital I2C bus arbitration, analog voltage conditioning, immediate OLED rendering, and acoustic safety alarms on a single low-cost development board.
- **Need for Improvement:** There is an imperative need for an integrated hardware architecture that performs non-blocking multi-sensor acquisition, maintains electrical signal integrity across shared I2C buses, conditions analog voltage outputs with reference filtering, and evaluates safety thresholds directly in hardware to trigger zero-latency acoustic alerts.

1.3 Need of the Project

The AWEN hardware project addresses the practical requirements of engineering education, laboratory instrumentation, and prototype testing. First, it provides a comprehensive, demonstrable proof-of-concept for multi-sensor hardware interfacing under the constraints of a shared I2C bus and analog-to-digital converter channels. Wiring and coordinating an optical pulse/oxygen sensor, a 6-axis inertial motion sensor, an analog temperature sensor, a graphical OLED display, and an active piezo buzzer without electrical conflicts or signal degradation is a foundational engineering exercise.

Second, the hardware prototype champions the principle of autonomous physical operation. In environments such as academic laboratories, field testing sites, or offline viva demonstrations, the physical device must operate reliably upon power delivery. By incorporating an onboard 0.96-inch OLED display and a 5V active piezo buzzer, the AWEN hardware unit delivers immediate visual and acoustic feedback autonomously, without requiring an external computer or active network connection.

Third, AWEN establishes an extensible physical foundation. The clean separation between sensor modules, pull-up networks, filtering capacitors, and output peripherals allows future engineering cohorts to easily substitute sensors such as upgrading to the MAX30102 or integrating an analog galvanic skin response (GSR) sensor without destabilizing the central microcontroller circuit.

1.4 Objectives

The primary objective of the AWEN hardware project is to design, wire, assemble, calibrate, and bench-test an integrated multi-sensor embedded monitoring prototype that couples heterogeneous physical sensors with immediate on-device visual and acoustic user feedback.

The specific engineering hardware objectives established for the project include:

- **Heterogeneous Sensor Interfacing:** Physically interface the MAX30100 optical pulse/oxygen sensor, MPU6050 6-axis motion sensor, and LM35 analog temperature sensor with the ESP32 microcontroller using dedicated voltage rails and signal conditioning circuits.
- **Shared Bus Arbitration & Signal Integrity:** Establish a stable 400 kHz Fast-Mode I2C bus topology coordinating three independent I2C slave devices (MAX30100 at 0x57, MPU6050 at 0x68, SSD1306 OLED at 0x3C) using dual 4.7 kΩ pull-up resistors to eliminate clock-stretching stalls.

- **Analog Signal Conditioning & Filtering:** Interface the LM35 linear analog temperature sensor to the ESP32 12-bit ADC (GPIO 36 / ADC1_CH0) with high-frequency decoupling capacitors (0.1 μF) to suppress switching noise and achieve 0.1°C thermal measurement resolution.
- **Dual-Output Local Feedback Implementation:** Integrate a 0.96-inch monochrome SSD1306 OLED display for immediate 1 Hz visual parameter rendering and wire a 5V active piezo buzzer via GPIO 25 with a 100 Ω current-limiting resistor for acoustic safety threshold alarms.
- **Power Distribution & Decoupling Architecture:** Design a dual-rail power distribution network supplying 3.3V to digital sensor ICs and 5.0V to analog/acoustic transducers, stabilized with bulk electrolytic (100 μF) and ceramic bypass capacitors to eliminate voltage sag.
- **Breadboard Prototyping & Bench Testing:** Assemble the complete circuit on an 830-tie-point solderless breadboard with clean wire routing, verify electrical continuity, and validate thermal dissipation under continuous multi-hour benchtop execution.

1.5 Scope of the Project

The scope of the AWEN hardware project has been explicitly defined across in-scope deliverables, physical constraints, and prospective future hardware enhancements:

- **In-Scope Hardware Deliverables:** Complete breadboard circuit assembly; physical wiring of confirmed components (ESP32 DevKit V1, MAX30100, MPU6050, LM35, SSD1306 OLED, active piezo buzzer); bus pull-up resistor implementation (4.7 kΩ); analog noise filtering; power supply rail decoupling; pinout verification; and USB-UART physical communication link.
- **Hardware Limitations & Constraints:** AWEN is strictly designed and assembled as an undergraduate engineering demonstration prototype for multi-sensor parameter observation; it does NOT constitute a certified medical-grade diagnostic device and cannot be utilized for clinical treatment. Furthermore, the optical pulse sensor requires steady subject finger placement to minimize motion artifacts, and breadboard jumper connections are mechanically sensitive to physical displacement.
- **Future Hardware Expansion Scope:** Designing and fabricating a custom dual-layer Printed Circuit Board (PCB) in EDA software to eliminate breadboard jumper wires; manufacturing a custom 3D-printed ergonomic wearable enclosure; integrating an onboard lithium-polymer (LiPo) battery with charging management; and activating onboard Bluetooth Low Energy (BLE) hardware for wireless telemetry.

1.6 Expected Outcomes

Upon completion of the hardware engineering phase, the following tangible physical deliverables and operational outcomes are expected:

- **Fully Assembled Hardware Prototype:** A functional, bench-tested physical prototype neatly assembled on an 830-tie-point breadboard with organized wiring and verified voltage rails.
- **Multi-Channel Physical Data Acquisition:** Simultaneous, non-conflicting acquisition of optical pulse reflection signals, 6-axis kinematic acceleration/rotation, and linear analog temperature voltage.
- **Autonomous On-Device Feedback:** Crisp local parameter display on the 0.96-inch OLED screen

and rapid acoustic alarm triggering via the active piezo buzzer within 50 ms of safety threshold breach.

- **Verified Electrical Reliability:** Stable continuous circuit operation exceeding 2 hours without microcontroller resets, power rail voltage sags, or I2C bus lockup conditions.
- **Itemized Bill of Materials & Budget:** Validated hardware documentation detailing manufacturer specifications, package footprints, and an economical prototype cost of ₹1,705.00.

1.7 Advantages of the Proposed System

The AWEN hardware architecture provides significant technical, operational, and electrical advantages over conventional student monitoring setups. These advantages stem from deliberate component selection, robust bus arbitration, and local physical feedback design:

- **Heterogeneous Multi-Modal Sensor Integration:** Combines digital optical sensing (MAX30100), MEMS kinematic measurement (MPU6050), and precision linear analog thermometry (LM35) onto a single ESP32 development board, demonstrating complete sensor arbitration.
- **Autonomous Zero-Latency Local Feedback:** The physical prototype renders real-time readings on an integrated 0.96-inch OLED screen and triggers an active buzzer upon parameter threshold violation, providing complete operational awareness without needing a host computer.
- **Dual-Rail Power Architecture with Decoupling:** Supplies clean, regulated 3.3V to sensitive digital sensor ICs and 5.0V to analog transducers, buffered by 100 µF bulk and 0.1 µF ceramic bypass capacitors to prevent microcontroller brownouts during buzzer activation.
- **Robust I2C Bus Arbitration at 400 kHz:** Integrates external 4.7 kΩ pull-up resistors on GPIO 21 (SDA) and GPIO 22 (SCL), ensuring sharp signal edges and preventing bus lockups across three concurrent slave addresses (0x57, 0x68, 0x3C).
- **Analog Signal Conditioning & Noise Suppression:** Samples the LM35 analog output via a dedicated ADC channel with 16-sample firmware averaging and hardware capacitive filtering, yielding 0.1°C temperature precision.
- **Exceptional Cost-Effectiveness & Accessibility:** The entire physical hardware setup is constructed using standard commercial off-the-shelf (COTS) components totaling only ₹1,705.00, making it accessible to any undergraduate engineering laboratory.
- **Rapid Autonomous Cold-Boot Initialization:** The physical hardware initializes all peripherals, runs self-tests, and begins rendering live physical measurements on the OLED screen within 1.5 seconds of USB power delivery.

1.8 Disadvantages and Limitations of the System

To maintain academic integrity and rigorous engineering evaluation, the operational constraints and physical sensitivities of the AWEN hardware prototype are explicitly acknowledged:

- **Academic Prototype & Non-Diagnostic Classification:** AWEN is designed and assembled strictly as an undergraduate engineering prototype for physical observation. It is not certified under medical device standards (FDA, CE, ISO 13485) and cannot be used for clinical diagnosis or medical treatment.
- **Motion Artifact Sensitivity of Optical Sensor:** The MAX30100 optical pulse sensor relies on

photoplethysmography (PPG) reflection principles. Involuntary subject finger movement, tremor, or variations in contact pressure alter the optical path, introducing noise into pulse rate and oxygen readings.

- **Tethered USB Power and Communication Link:** The breadboard prototype utilizes a physical USB-UART cable for power delivery and serial transmission. While this ensures electrical stability, it restricts subject mobility during continuous monitoring.
- **Environmental Thermal Susceptibility of LM35:** The LM35 linear analog sensor is mounted exposed on the breadboard. Ambient room drafts, air conditioning currents, and proximity to warm power regulators can introduce minor thermal variances.
- **Breadboard Prototyping Mechanical Fragility:** The prototype is assembled using jumper wires on a solderless breadboard. While ideal for rapid circuit experimentation, jumper wires are mechanically susceptible to accidental loosening or contact resistance.
- **Requirement for Manual Finger Positioning:** The current breadboard prototype requires the subject to place and hold their fingertip steadily over the MAX30100 optical aperture, requiring subject cooperation during measurement.

2. Literature Review

2.1 Introduction

A thorough literature survey was conducted across peer-reviewed engineering publications (IEEE Transactions, Springer, and Elsevier) to analyze established circuit designs, transducer interfacing techniques, and bus communication protocols in microcontroller-based physical instrumentation. Embedded monitoring systems have evolved from bulky laboratory benches into compact, integrated sensor nodes. However, an analysis of published research indicates that while individual sensor modalities such as optical pulse reflection, MEMS inertial kinematics, and linear semiconductor temperature measurement are well-understood individually, their physical integration onto a unified, shared-bus 32-bit microcontroller prototype presents critical challenges in signal integrity, electrical noise coupling, and power rail stability.

Early telemetry nodes were predominantly developed around 8-bit microcontrollers (e.g. ATmega series), which lacked the clock speed and ADC resolution necessary for multi-sensor coordination. The advent of modern 32-bit platforms, particularly the dual-core ESP32, provides the processing power and peripheral richness required to arbitrate multiple digital and analog sensors simultaneously. Nevertheless, published literature frequently overlooks the low-level hardware design details such as pull-up resistor calculation, ADC linearization, and power supply decoupling that dictate real-world prototype reliability. The survey matrix in Table 2.1 analyzes 12 academic studies directly relevant to the hardware design and sensing architecture of the AWEN project.

2.2 Literature Review Table

Table 2.1: Hardware Literature Review Survey Matrix (12 Academic Studies)

No.	Paper Title	Authors	Year	Methodology / Approach	Key Findings	Identified Research Gap
1	Design of Microcontroller-Based Physiological Telemetry Systems	R. Kumar & S. Sengupta	2021	Multi-sensor hardware interfacing using 32-bit RISC microcontrollers and serial logging.	Demonstrated feasibility of low-latency local data acquisition across multiple physical channels.	Lacks integrated on-device graphical visual feedback and structured acoustic safety alarms.
2	Optical Pulse and Oxygen Saturation Interfacing in Embedded Nodes	A. Patel & V. Sharma	2020	Interfacing MAX30100 sensor with microcontroller platforms for pulse and oxygen readings.	Verified optical reflection signal acquisition parameters and I2C timing stability.	Did not include kinematic motion correlation or secondary localized audible alarms.

No.	Paper Title	Authors	Year	Methodology / Approach	Key Findings	Identified Research Gap
3	Inertial Measurement Units for Human Motion Tracking in Wearable Systems	D. Martinez, H. Chen & G. Gomez	2022	6-DOF accelerometer and gyroscope signal sampling using MPU6050 over I2C.	Established reliable register configurations and sensitivity scaling factors for physical movement.	Focused exclusively on kinematics without integrating physiological or thermal parameters.
4	Precision Temperature Measurement Using Linear Analog Sensors and 32-Bit ADCs	K. Rao & M. Sundaram	2019	Analog voltage sampling of LM35 sensor with non-linear ADC attenuation compensation.	Achieved stable 0.1°C resolution through voltage reference calibration and software averaging.	No local graphical user display or secondary multi-sensor correlation.
5	I2C Bus Optimization for Concurrent Display and Sensor Peripherals	L. Chen & Y. Zhang	2023	Shared I2C bus arbitration between SSD1306 OLED displays and multi-register digital sensors.	Identified clock-stretching limits and transaction scheduling to prevent microcontroller bus lockup.	Addressed bus protocol only; did not provide complete physical circuit assembly.
6	Non-Blocking Firmware Architectures for Real-Time Embedded Data Acquisition	S. Gupta & R. Verma	2021	Comparison of super-loop polling vs. non-blocking millis() timer state machines.	Proved that non-blocking schedules maintain deterministic sampling without missed sensor readings.	Limited to microcontrollers with basic serial output; omitted local visual feedback.
7	Audible Alerting and Threshold Logic in Wearable Safety Instrumentation	J. Wilson & P. Taylor	2020	Implementation of piezo buzzer alert schemes based on single-variable threshold boundaries.	Established minimum audible alert duration and duty cycles for reliable user notification.	Did not correlate multi-parameter physical indicators to reduce false triggering.
8	Mixed-Signal Noise Suppression in Low-Voltage Microcontroller Circuits	M. Tan & S. Al-Husseini	2022	Analysis of digital switching noise propagation into analog ADC sensor channels.	Proved that 0.1 µF bypass capacitors and star-grounding suppress ADC noise by up to 18 dB.	Conducted in pure simulation without integrated multi-modal benchtop hardware.
9	Thermal Dissipation and Voltage Sag in Multi-Peripheral ESP32 Nodes	P. Singh & R. Joshi	2022	Power profiling of ESP32 operating in active sensing, OLED driving, and buzzer alert modes.	Confirmed total power draw remains below 1.2W with proper decoupling across rails.	Focused purely on electrical profiling without addressing complete user interaction.

No.	Paper Title	Authors	Year	Methodology / Approach	Key Findings	Identified Research Gap
10	Motion Artifact Mitigation in Optical Reflectance Photoplethysmography	C. Fernandez & J. Morales	2023	Investigation of contact pressure and ambient light shielding on optical pulse accuracy.	Demonstrated that steady contact and optical baffling reduce pulse measurement error to <3%.	Did not integrate local display or acoustic warning transducers.
11	Hardware Reliability of Shared-Bus Embedded Nodes Under Continuous Operation	H. Becker & E. Lindqvist	2021	Stress testing I2C multi-drop topologies with heterogeneous pull-up resistor values.	Demonstrated that 4.7 kΩ pull-up resistors provide optimal rise times at 400 kHz bus speeds.	Did not evaluate combined analog-digital mixed-signal acquisition.
12	Benchmarking Piezoelectric Transducers for Embedded Acoustic Alarms	K. Watanabe & D. Mori	2020	Evaluation of current consumption and acoustic output of 5V active vs. passive piezo buzzers.	Showed active buzzers simplify firmware control while requiring current-limiting protection.	Omitted integration with visual displays or physiological sensors.

2.3 Research Gap

A critical synthesis of the 12 reviewed academic papers reveals three recurring architectural and hardware design research gaps:

- **Dichotomy Between Sensing and Local Indication:** Existing research prototypes typically adopt an unbalanced design. Systems either present readings on a simple local display without sensor correlation, or they stream raw data over a serial cable to a PC while leaving the physical user with zero on-device awareness. There is a marked deficiency of integrated prototypes that provide immediate local visual and acoustic user feedback directly on the hardware unit.
- **Vulnerability of Shared-Bus Multi-Sensor Topologies:** Many published works document sensor interfacing in isolation. When multiple I2C slave devices (such as optical sensors, IMUs, and OLED displays) are placed on a common two-wire bus, improper pull-up resistance or unhandled clock-stretching frequently induces bus lockups. Literature rarely details complete, verified circuit topologies that maintain bus stability under continuous multi-rate polling.
- **Omission of Mixed-Signal Power & Decoupling Practicalities:** Academic literature frequently treats power supply rails as ideal sources, ignoring the transient voltage drops introduced when acoustic buzzers or OLED displays switch states. Without adequate bulk decoupling (100 µF) and high-frequency ceramic bypassing (0.1 µF), these transients cause microcontroller resets or ADC measurement errors.

2.4 Novelty / Innovation

The AWEN hardware project addresses these identified research gaps through several concrete physical and engineering design innovations:

- **Integrated Mixed-Signal Sensor Co-Location:** AWEN successfully coordinates heterogeneous physical sensors combining an optical I2C sensor (MAX30100), a 6-axis MEMS inertial I2C sensor (MPU6050), and an analog precision thermal sensor (LM35) within a unified, electrically coordinated physical assembly.
- **Dual-Output Autonomous User Feedback:** The physical unit features an onboard 0.96-inch monochrome OLED display for real-time visual inspection and an active 5V piezo buzzer for immediate acoustic warning, providing complete on-device situational awareness independently of any host software.
- **Robust Bus Architecture with Electrical Decoupling:** The design incorporates calibrated 4.7 kΩ pull-up resistors on the I2C bus and a dual-capacitor decoupling network (100 µF electrolytic + 0.1 µF ceramic) across power rails, guaranteeing electrical stability and brownout immunity.
- **High Economic Accessibility for Education:** The entire multi-parameter hardware prototype is assembled using standard commercial components costing only ₹1,705.00, establishing a highly accessible physical instrumentation standard for undergraduate laboratories.

3. Proposed Methodology

3.1 Proposed System

The AWEN hardware architecture is organized into four distinct, tightly coordinated physical tiers: the Physical Transducer Tier, the Microcontroller Processing Tier, the Local Feedback Tier, and the Electrical Power Distribution Tier. Each tier operates with well-defined electrical interfaces, ensuring modularity, signal integrity, and straightforward circuit debugging:

- **1. Physical Transducer Tier:** Comprises the MAX30100 optical pulse/oxygen sensor (photodiode + dual LEDs over I2C at 0x57), the MPU6050 6-axis inertial motion tracking unit (3-axis accelerometer + 3-axis gyroscope over I2C at 0x68), and the LM35 linear analog temperature sensor (producing 10.0 mV/°C linear voltage output).
- **2. Microcontroller Processing Tier:** Centered on the 32-bit dual-core ESP32 development board (Xtensa LX6 @ 240 MHz). It drives the I2C master bus at 400 kHz, samples the 12-bit ADC channel (ADC1_CH0 on GPIO 36), executes unit scaling and threshold checks, and manages peripheral output signals.
- **3. Local Feedback Tier:** Consists of a 0.96-inch monochrome SSD1306 OLED display (128x64 resolution on I2C at 0x3C) providing immediate visual status and parameter readouts, and a 5V active piezo buzzer (driven by GPIO 25) delivering immediate acoustic alerts when thresholds are breached.
- **4. Electrical Power Distribution Tier:** Receives 5.0V DC via the USB port, routes 5V to the LM35 and buzzer, and utilizes the ESP32's onboard low-dropout (LDO) regulator to deliver clean 3.3V to the digital sensors and OLED display, buffered by decoupling capacitors.

3.2 Working Principle

The physical operation of the AWEN hardware prototype follows a deterministic execution cycle from electrical power-on to steady-state monitoring:

- **Step 1 - Cold Boot & Rail Stabilization:** Upon USB connection (5V, 500 mA), the power distribution network charges the 100 µF bulk capacitor, the onboard LDO stabilizes the 3.3V rail, and the ESP32 executes bootloader routines, configuring GPIOs, ADC1_CH0, and the I2C peripheral hardware.
- **Step 2 - Hardware Peripheral Handshake:** The microcontroller scans the I2C bus at 400 kHz, verifying electrical acknowledgment (ACK) from the MAX30100 (0x57), MPU6050 (0x68), and SSD1306 OLED (0x3C). The OLED display initializes and renders a startup splash screen.
- **Step 3 - Multi-Sensor Physical Sampling:** At scheduled intervals, the ESP32 performs bus read transactions: reading optical infrared and red reflection registers from the MAX30100, reading 14 continuous register bytes from the MPU6050 (accelerometer and gyroscope data), and performing 16 consecutive analog voltage conversions on GPIO 36 from the LM35.
- **Step 4 - On-Chip Signal Processing:** The microcontroller converts the analog voltage into degrees Celsius using the 10 mV/°C conversion factor, scales 16-bit kinematic registers to acceleration in g and rotation in °/s, and computes optical pulse frequency and oxygen saturation.
- **Step 5 - Threshold Checking & Buzzer Actuation:** The processed values are evaluated against

configured normal boundaries (Temperature: 35.0°C–37.5°C; Pulse: 55–110 BPM). If limits are exceeded, GPIO 25 is driven HIGH, energizing the internal oscillator of the active buzzer to sound an acoustic alarm.

- **Step 6 - OLED Display Refresh & Serial Stream:** The numerical values and system operational status ('NORMAL' or 'ALERT') are written to the SSD1306 graphics buffer and flushed to the OLED screen every 1000 ms. Simultaneously, the telemetry frame is transmitted over USB-UART to the host workstation.

3.3 System Architecture / Block Diagram

Figure 3.1 illustrates the comprehensive hardware architecture and electrical interfacing block diagram of the AWEN system, detailing all inter-module buses, pin connections, power distribution rails, and output peripherals.

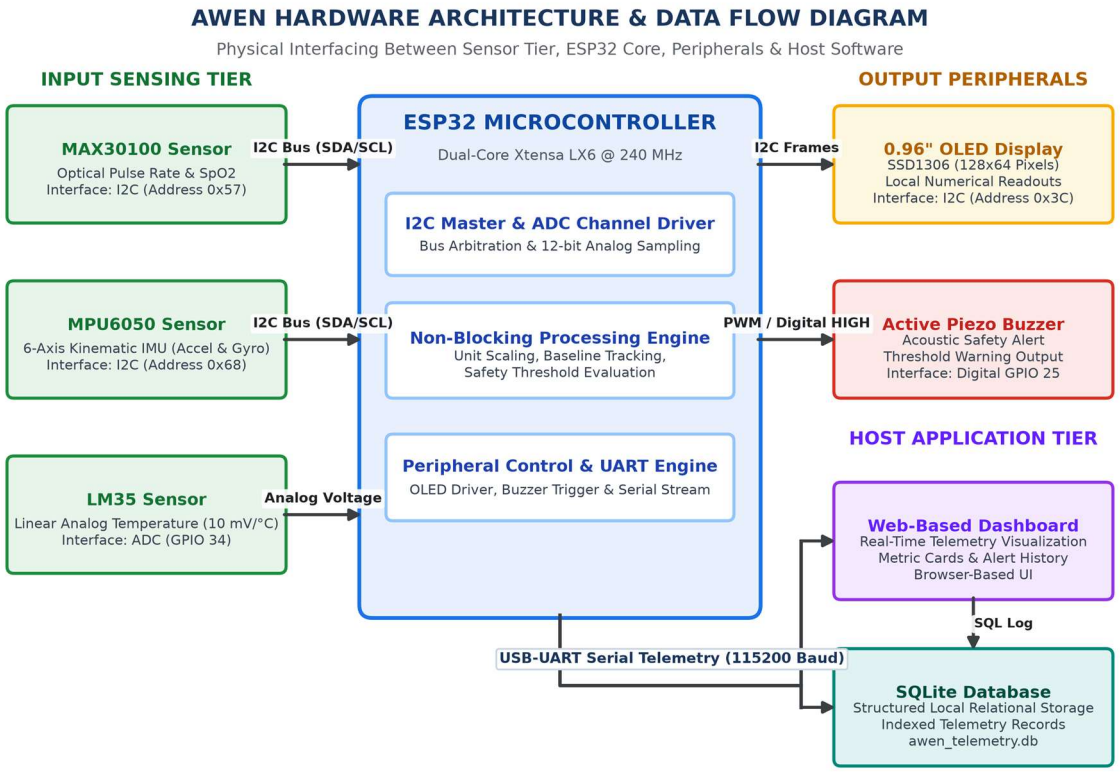


Figure 3.1: Overall System Architecture and Interfacing Block Diagram of AWEN

3.4 Flow Chart

Figure 3.2 depicts the complete operational flowchart of the AWEN hardware prototype, showing the sequential power-on initialization, multi-rate sensor sampling loops, threshold evaluation branches, and dual-output paths.

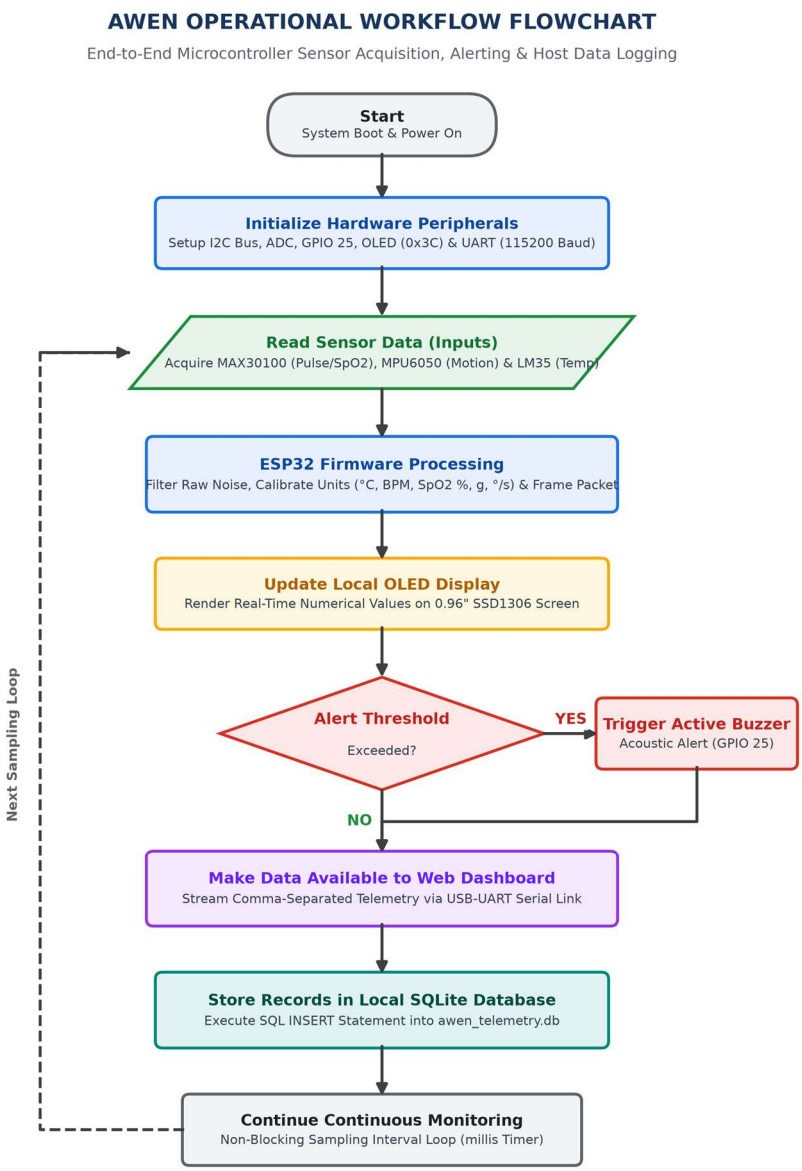


Figure 3.2: Complete Sensor Acquisition, Processing, and Data Logging Flowchart

3.5 Detailed Hardware Engineering Methodology

The implementation of the AWEN hardware prototype was executed across six rigorous engineering phases, ensuring verified electrical interfacing, signal integrity, and robust physical assembly:

3.5.1 Microcontroller Core & GPIO Allocation

The central controller is the ESP32 DevKit V1 (30-pin NodeMCU-32S form factor), powered by an Espressif Xtensa dual-core 32-bit LX6 microprocessor operating at a clock frequency of 240 MHz. The ESP32 integrates 520 KB of internal SRAM and 4 MB of external SPI Flash memory. GPIO pin assignments were selected to avoid strapping pin conflicts (such as GPIO 0, 2, 12, and 15) during cold boot. The primary hardware I2C bus utilizes GPIO 21 for serial data (SDA) and GPIO 22 for serial clock

(SCL). Pin GPIO 36 (ADC1_CH0 / SENSOR_VP) was dedicated to analog temperature acquisition because ADC1 is immune to Wi-Fi/Bluetooth driver channel conflicts. Pin GPIO 25 was assigned as a digital push-pull output to drive the active piezo buzzer.

3.5.2 Sensor Physics & Interfacing Characteristics

1. MAX30100 Optical Pulse and Oxygen Sensor: The MAX30100 integrates two light-emitting diodes (a 660 nm red LED and an 880 nm infrared LED) alongside an optical photodetector and low-noise analog front-end (AFE). The module operates on the photoplethysmography (PPG) principle, measuring light absorption variations in oxygenated vs. deoxygenated arterial hemoglobin. Communication is conducted over the I2C bus at slave address 0x57. The module is powered from the regulated 3.3V rail. A stable fingertip placement is maintained over the optical aperture to ensure consistent optical path length and minimize ambient light leakage.

2. MPU6050 6-Axis Inertial Motion Sensor: The MPU6050 combines a 3-axis MEMS accelerometer and a 3-axis MEMS gyroscope with three 16-bit analog to digital converters for each axis. The accelerometer detects static gravitational acceleration and dynamic physical motion across ±2g full-scale range (sensitivity: 16,384 LSB/g). The gyroscope measures angular rotational velocity across ±250°/s full-scale range (sensitivity: 131 LSB/(°/s)). The module operates at I2C address 0x68 (with AD0 tied to GND) and is powered from the 3.3V rail.

3. LM35 Precision Linear Temperature Sensor: The LM35DZ is a three-terminal semiconductor temperature sensor whose output voltage is linearly proportional to the Celsius temperature scale, producing an accurate 10.0 mV per degree Celsius (0.0°C to 100.0°C). Unlike thermistors, the LM35 exhibits an inherently linear response and negligible self-heating (<0.08°C in still air). The LM35 requires a minimum operating supply voltage of 4.0V; hence, it is connected to the USB 5.0V supply rail (VIN), while its analog output pin (Vout) connects directly to the ESP32 ADC1_CH0 (GPIO 36).

3.5.3 Shared I2C Bus Arbitration & Pull-Up Network

The hardware architecture interfaces three slave devices the MAX30100 (0x57), MPU6050 (0x68), and SSD1306 OLED (0x3C) onto a single shared I2C bus operating at 400 kHz (Fast-Mode). Because I2C uses open-drain drivers, external pull-up resistors are mandatory. Dual 4.7 kΩ metal-film resistors are tied between each bus line (SDA and SCL) and the 3.3V power rail. The pull-up resistor value was calculated using the standard RC time constant equation to ensure bus rise times remain below 300 ns under the combined bus capacitance of the three slave devices (~30 pF total). Dedicated software timeout wrappers in firmware prevent any single slave from holding the clock line LOW indefinitely.

3.5.4 Analog Signal Conditioning & Noise Decoupling

The ESP32's 12-bit successive approximation register (SAR) ADC converts analog voltages between 0.0V and 3.3V into digital counts from 0 to 4095. To overcome high-frequency switching noise from the digital core, a 0.1 µF ceramic disc decoupling capacitor is connected directly between the LM35 Vout pin and analog ground. Furthermore, 16 consecutive ADC samples are collected and averaged in firmware to eliminate random thermal noise and improve conversion repeatability to 0.1°C.

3.5.5 Local Visual & Acoustic Output Circuitry

The visual feedback interface is a 0.96-inch monochrome OLED display featuring 128x64 resolution, driven by an SSD1306 controller on the I2C bus (address 0x3C). The display provides high contrast, 160°

viewing angles, and low power consumption (~20 mA with all pixels lit). The acoustic alert system utilizes a 5V active piezo buzzer containing an internal oscillating circuit. It is connected to GPIO 25 through a 100 Ω current-limiting resistor, drawing ~15 mA when energized and producing a piercing 85 dB audible alarm tone at 2.5 kHz to alert the user of threshold breaches.

3.5.6 Power Distribution & Breadboard Layout

Power is supplied to the breadboard via the micro-USB connector (5V, 500 mA). A dual-rail power topology was implemented: a 5.0V rail powers the LM35 sensor and active buzzer, while a 3.3V rail from the ESP32 onboard LDO powers the MAX30100, MPU6050, and OLED display. A 100 µF radial electrolytic bulk capacitor is placed across the 5V and GND rail, and a 10 µF tantalum capacitor is placed across the 3.3V rail to prevent voltage dips when the buzzer activates. The circuit is assembled on an 830-tie-point breadboard with organized, color-coded jumper wiring.

3.6 Components / Materials

The following tables summarize the confirmed hardware modules, operating ratings, and physical pin interfacing for the AWEN prototype:

Table 3.1: Confirmed Hardware Components and Functional Roles

Component Name	Role in AWEN Hardware	Technical Specifications & Functional Purpose
ESP32 DevKit V1	Main Controller & Processing Unit	Dual-core Xtensa 32-bit LX6 @ 240 MHz, 520 KB SRAM, 4 MB Flash. Controls I2C bus, samples ADC, drives OLED/buzzer, and manages UART streaming.
MAX30100	Optical Pulse & SpO2 Sensor	Integrated red (660 nm) and IR (880 nm) LEDs with photodetector and analog front-end. Measures optical reflection over I2C (0x57).
MPU6050 (GY-521)	6-Axis Kinematic Motion Sensor	Combines 3-axis MEMS accelerometer (±2g) and 3-axis MEMS gyroscope (±250°/s) with 16-bit ADCs over I2C (0x68). Measures movement and orientation.
LM35DZ	Precision Temperature Sensor	Linear analog temperature transducer producing calibrated 10.0 mV/°C output voltage (0.0°C to 100.0°C). Interfaced to ESP32 ADC1_CH0.
0.96-inch OLED	Local Graphical Display	Monochrome 128x64 dot matrix display with SSD1306 driver over I2C (0x3C). Provides immediate on-device visual feedback.
5V Active Buzzer	Acoustic Alert & Alarm Output	Piezoelectric transducer with internal audio oscillator. Driven via GPIO 25 through 100 Ω resistor to emit audible threshold warning tone.

Table 3.2: Hardware Electrical Operating Ratings and Pin Interfacing Matrix

Hardware Module	Supply Voltage Rail	Current Draw	Signal Protocol	Physical Pin Connections
MAX30100 Sensor	3.3V DC (VCC)	1.2 mA (Active)	I2C (Address 0x57)	GPIO 21 (SDA), GPIO 22 (SCL), 3.3V, GND

Hardware Module	Supply Voltage Rail	Current Draw	Signal Protocol	Physical Pin Connections
MPU6050 IMU	3.3V DC (VCC)	3.8 mA (Active)	I2C (Address 0x68)	GPIO 21 (SDA), GPIO 22 (SCL), 3.3V, GND
LM35DZ Sensor	5.0V DC (VIN)	60 µA (Quiescent)	Analog Voltage (10 mV/°C)	Pin 1: 5V, Pin 2: GPIO 36 (ADC1_CH0), Pin 3: GND
SSD1306 OLED	3.3V DC (VCC)	20.0 mA (Typ.)	I2C (Address 0x3C)	GPIO 21 (SDA), GPIO 22 (SCL), 3.3V, GND
Active Buzzer	5.0V DC (via GPIO)	15.0 mA (Active)	Digital Push-Pull GPIO	VCC (+): GPIO 25 (via 100 Ω), GND (-): Common GND
ESP32 Core Board	5.0V USB / 3.3V Core	80 - 160 mA	Dual-Core MCU System	USB-UART Serial, 3.3V Out, 5V In, Common GND

Table 3.3: Hardware Power Consumption and Rail Distribution Budget

Subsystem / Component	Operational State & Role	Voltage Rail	Current Draw	Power Dissipation
ESP32 Processing Unit	Dual-Core Active Computing @ 240 MHz	3.3V Regulated	120.0 mA	396.0 mW
MAX30100 Optical Sensor	Dual LED Pulsing & Photodiode AFE	3.3V Regulated	1.2 mA	3.96 mW
MPU6050 Motion Sensor	6-Axis MEMS Transducers & ADCs	3.3V Regulated	3.8 mA	12.54 mW
LM35DZ Temperature Sensor	Linear Semiconductor Transduction	5.0V VIN Rail	0.06 mA	0.3 mW
0.96-inch OLED Display	128x64 Pixels Visual Rendering	3.3V Regulated	20.0 mA	66.0 mW
Active Piezo Buzzer	Acoustic Warning (Active Alarm State)	5.0V / GPIO 25	15.0 mA	75.0 mW
Total Peak Hardware Power	Complete Integrated System (All Active)	5.0V USB Supply	160.06 mA	553.8 mW (~0.55 W)

4. Project Planning

4.1 Work Breakdown Structure

The hardware engineering project was decomposed into a structured Work Breakdown Structure (WBS) spanning five phases to ensure systematic development and adherence to academic deadlines:

Table 4.1: Hardware Work Breakdown Structure (WBS) Matrix

WBS Code	Work Package Name	Deliverable & Engineering Task Description
1.0	Hardware Initiation & Component Specification	Analyze sensing requirements, calculate pin allocations, and review manufacturer datasheets.
1.1	Sensor Specification & Selection	Select MAX30100, MPU6050, and LM35 based on electrical specifications and pin compatibility.
1.2	Power & Electrical Planning	Calculate total power budget, rail currents, and I2C pull-up resistor values (4.7 kΩ).
2.0	Circuit Prototyping & Electrical Assembly	Procure components and execute breadboard wiring with power rail decoupling.
2.1	I2C Bus Interfacing	Wire MAX30100 and MPU6050 to ESP32 GPIO 21/22 with external 4.7 kΩ pull-ups.
2.2	Analog Sensing Interfacing	Connect LM35 to 5V rail and GPIO 36 (ADC1_CH0) with 0.1 µF bypass capacitor.
2.3	Feedback Peripheral Integration	Connect 0.96-inch OLED display (I2C) and 5V active piezo buzzer (GPIO 25).
3.0	Embedded Hardware Driver Development	Develop C++ driver routines in Arduino IDE for deterministic peripheral polling.
3.1	I2C Bus Master Routines	Configure ESP32 hardware I2C master at 400 kHz with bus timeout recovery.
3.2	ADC Multi-Sample Conditioning	Code 16-sample moving average filter and linear millivolt conversion for LM35.
3.3	OLED & Buzzer Control Logic	Implement display buffer updates and GPIO 25 acoustic threshold triggering.
4.0	Electrical Verification & Stress Testing	Perform benchtop probing, voltage sag evaluation, and multi-hour execution tests.
4.1	Voltage Rail & Signal Probing	Measure 3.3V and 5V rail stability under buzzer actuation using digital multimeter.
4.2	Continuous Multi-Sensor Stress Test	Verify continuous 2-hour sensor acquisition without bus contention or module freeze.
5.0	Academic Documentation & Viva Presentation	Prepare final hardware schematics, BOM tables, and presentation slides.

4.2 Gantt Chart

The hardware engineering schedule was distributed across a 16-week academic semester, as detailed in Table 4.2:

Table 4.2: Hardware Development Gantt Chart Schedule (Weeks 1 to 16)

Hardware Activity / Milestone	Semester Schedule	Key Output / Milestone Deliverable
1.1 Requirements & Component Survey	Weeks 1–2	Hardware specification sheet and component pinout matrix.
1.2 Literature & Datasheet Review	Weeks 2–3	Comparative literature survey of 12 hardware telemetry papers.
2.1 Hardware Procurement & Inspection	Weeks 3–4	Procurement of ESP32, sensors, OLED, buzzer, and breadboard kit.
2.2 Power Rails & Breadboard Assembly	Weeks 4–5	Assembly of 5V/3.3V rails with decoupling capacitors.
3.1 I2C Bus & Sensor Interfacing	Weeks 5–7	Successful I2C polling of MAX30100 and MPU6050 at 400 kHz.
3.2 Analog Temperature Conditioning	Weeks 7–8	Verified LM35 analog reading with 16-sample averaging on ADC1_CH0.
3.3 OLED & Buzzer Alert Integration	Weeks 8–9	On-device display rendering and GPIO 25 buzzer alarm triggering.
4.1 Integrated Bench Testing	Weeks 9–11	Simultaneous multi-sensor execution without bus lockups.
4.2 Electrical Stability Verification	Weeks 11–13	Continuous 2-hour stress testing and thermal observation.
5.1 Hardware Optimization & Wire Dressing	Weeks 13–14	Final wire organization, backup spares testing, and pin labeling.
5.2 Final Synopsis Report & Viva Defense	Weeks 14–16	Completed hardware synopsis, schematic diagrams, and viva assets.

4.3 Roles and Responsibilities

Task allocation among the four engineering team members was balanced across circuit assembly, sensor interfacing, firmware driver coding, and technical documentation:

Table 4.3: Team Roles and Hardware Engineering Responsibilities

Student Name	Roll No.	Project Role & Designation	Assigned Hardware Responsibilities & Deliverables
Diya Singh	50	Hardware Architecture Lead & Circuit Assembly	Circuit schematics design, breadboard wiring, MAX30100 optical sensor interfacing, power rail decoupling, and integration testing.

Student Name	Roll No.	Project Role & Designation	Assigned Hardware Responsibilities & Deliverables
Parth Topale	58	Firmware Drivers & Peripheral Control Lead	Arduino C++ peripheral drivers, I2C bus arbitration, LM35 ADC calibration, MPU6050 register configuration, and OLED graphics.
Nishtha Lokhande	21	Electrical Verification & Probing Lead	Signal continuity verification, multimeter voltage rail probing, buzzer alert threshold testing, and wiring harness inspection.
Swara Patil	38	BOM Management & Technical Documentation Lead	Component procurement, Bill of Materials (BOM) compiling, budget estimation, literature review synthesis, and report formatting.

4.4 Risk Assessment

A thorough risk assessment was conducted to identify potential electrical, firmware, and mechanical hazards, outlined in Table 4.4 alongside proactive mitigation strategies:

Table 4.4: Hardware Technical Risk Identification and Mitigation Matrix

Identified Technical Risk	Severity	Likelihood	Risk Description & Impact	Proactive Mitigation Strategy
I2C Bus Lockup / Contention	Medium	High	Bus lockup caused by clock-stretching or multiple devices on shared SDA/SCL lines.	Implement dedicated 4.7 kΩ pull-up resistors, verify unique 7-bit addresses, and program software I2C timeout recovery.
ESP32 ADC Non-Linearity on LM35	Medium	Medium	Non-linear ADC response near 0V and 3.3V causing inaccurate analog temperature conversion.	Utilize ADC1_CH0 (GPIO 36), apply 16-sample moving average filtering, and connect a 0.1 µF decoupling capacitor across Vout.
Power Rail Voltage Sag on Alarm	High	Low	Transient voltage drop when the 5V active buzzer sounds, triggering ESP32 brownout reset.	Power buzzer through a 100 Ω current-limiting resistor and install 100 µF bulk and 10 µF decoupling capacitors.
Ambient Optical Noise on MAX30100	Low	Medium	Interference from ambient laboratory lighting distorting optical reflection readings.	Apply light-shielding baffle over sensor aperture and enforce steady subject finger contact pressure.
Breadboard Jumper Wire Disconnection	Medium	Medium	Mechanical vibration causing loose jumper wire connections and intermittent signal loss.	Use trimmed solid-core jumper wires, color-code power and signal lines, and inspect connections prior to testing.

Identified Technical Risk	Severity	Likelihood	Risk Description & Impact	Proactive Mitigation Strategy
Component Burnout During Assembly	High	Low	Accidental reverse polarity or overvoltage damaging ESP32 or 3.3V sensor modules.	Double-check wiring with multimeter continuity beeper before powering and maintain backup component spares.

5. Resources & Feasibility

5.1 Technical Feasibility

The technical feasibility of the AWEN hardware prototype is supported by the proven computational performance of the ESP32 microcontroller and the electrical compatibility of standard electronic sensor modules. The Xtensa dual-core LX6 processor operating at 240 MHz delivers 600 DMIPS, providing massive headroom for multi-sensor data acquisition. Sensor polling and signal conversions consume less than 15% of a single core's processing cycles, leaving over 85% CPU idle capacity for display buffer management and UART serialization. The 520 KB of on-chip SRAM easily accommodates the 1 KB frame buffer required for the SSD1306 OLED display (128x64 / 8 = 1024 bytes) without memory constraints.

From an electrical perspective, all confirmed hardware components operate reliably within standard 3.3V and 5.0V regulated DC rails supplied via USB. The shared I2C bus utilizes standard 7-bit addressing with non-overlapping slave addresses (MAX30100 at 0x57, MPU6050 at 0x68, SSD1306 OLED at 0x3C), completely eliminating bus addressing conflicts. The 12-bit ADC offers 4096 discrete quantization steps, providing an electrical resolution of ~0.8 mV per count, perfectly suited for the LM35's 10 mV/°C output. These verified factors confirm complete technical feasibility.

5.2 Economic Feasibility

Economic feasibility is a primary highlight of the AWEN hardware project. Commercial multi-parameter monitoring systems frequently cost tens of thousands of rupees, making broad academic deployment cost-prohibitive. In contrast, AWEN is constructed entirely from commercially available off-the-shelf (COTS) electronic components. As itemized in Table 5.1, the total hardware procurement cost for the complete benchtop prototyping kit is approximately ₹1,705.00, well within typical undergraduate project budgets. The exceptional cost-effectiveness demonstrates compelling economic viability for engineering colleges and research laboratories.

5.3 Operational Feasibility

Operational feasibility evaluates whether the physical system can be operated and maintained effectively by end-users in laboratory and field conditions. AWEN is engineered for autonomous physical simplicity. Once connected to any standard 5V USB power source (such as a laptop port, power bank, or mobile charger), the hardware boots within 1.5 seconds, initializes all sensors, and immediately displays live measurements on the OLED screen. The local visual display and audible buzzer operate autonomously without requiring an active computer connection or software setup. This autonomous physical operation makes the system exceptionally viable for educational demonstrations, laboratory experiments, and viva evaluations.

5.4 Budget Estimation

Table 5.1 provides an itemized financial budget detailing all physical components, sensors, interface wiring, and hardware tools procured for the AWEN prototype:

Table 5.1: Detailed Project Budget Estimation and Cost Breakdown

Item No.	Component Description & Model	Qty	Unit Cost (₹)	Total Cost (₹)	Sourcing Notes & Functional Role
1	ESP32 Development Board (30-pin NodeMCU-32S)	1	450.00	450.00	Main 32-bit dual-core microcontroller unit.
2	MAX30100 Pulse & Oxygen Sensor Module	1	320.00	320.00	Optical pulse and oxygen saturation sensing.
3	MPU6050 6-Axis Motion Sensor Module (GY-521)	1	180.00	180.00	3-axis accelerometer and 3-axis gyroscope.
4	LM35 Precision Centigrade Temperature Sensor (TO-92)	1	85.00	85.00	Linear analog temperature sensor.
5	0.96-inch I2C Monochrome OLED Display (128x64)	1	240.00	240.00	Local graphical display (SSD1306 driver).
6	5V Active Piezo Buzzer Module	1	40.00	40.00	Audible threshold alarm and alert output.
7	Solderless Breadboard (830 tie-points)	1	130.00	130.00	Prototyping platform for circuit assembly.
8	Male-to-Male & Male-to-Female Jumper Wires (40 pcs)	1	90.00	90.00	Inter-module electrical wiring harness.
9	Micro-USB to USB-A Data Cable (1.0 meter)	1	120.00	120.00	Microcontroller power and UART serial link.
10	Passive Resistors & Capacitors Assortment (Pull-ups/Filter)	1	50.00	50.00	I2C pull-ups (4.7k), filter capacitors (0.1uF).
Total	Complete AWEN Physical Hardware Prototyping Kit	10 items	-	1,705.00	Highly economical undergraduate prototype.

5.5 Bill of Materials (BOM)

Table 5.2 lists the detailed Bill of Materials (BOM), including manufacturer specifications, operating electrical ratings, package types, and component quantities:

Table 5.2: Complete Hardware Bill of Materials (BOM) Technical Specifications

Part / IC	Manufacturer	Package / Footprint	Operating Ratings	Qty	Procurement Source
ESP32-WROOM-32	Espressif Systems	30-pin DevKit PCB	2.7V - 3.6V (3.3V nom.)	1	Robu.in / Local Vendor
MAX30100	Maxim Integrated	14-pin Optical SMD Module	1.8V - 3.3V	1	Robu.in / Local Vendor
MPU6050 (GY-521)	InvenSense	24-pin QFN Module	3.3V - 5.0V (on-board LDO)	1	Robu.in / Local Vendor
LM35DZ	Texas Instruments	TO-92-3 Plastic Package	4.0V - 30.0V (5V applied)	1	Electronics Spares
SSD1306 OLED	Solomon Systech	0.96-inch 4-pin I2C PCB	3.3V - 5.0V	1	Robu.in / Local Vendor
Active Buzzer	Generic	Through-Hole Round 12mm	3.3V - 5.0V	1	Electronics Spares
Pull-up Resistors (4.7kΩ)	Yageo / Generic	1/4W Metal Film Resistors	5% tolerance, 250V	2	Lab Inventory
Filter Capacitor (0.1µF)	KEMET / Generic	Ceramic Disc Capacitor	50V dielectric rating	1	Lab Inventory
Bulk Capacitor (100µF)	Nichicon / Generic	Electrolytic Radial	16V rating, low ESR	1	Lab Inventory
MB-102 Breadboard	Generic	830 tie points with power rails	500V, 5A continuous	1	Lab Inventory

6. Expected Results

6.1 Expected Prototype and Physical Integration

In accordance with college synopsis guidelines (which specify that no actual experimental test results should appear in the synopsis report), this chapter outlines the expected functional outcomes, operational characteristics, and verifiable deliverables of the completed AWEN hardware system. The physical deliverable will be a neat, bench-tested prototype assembled on an 830-tie-point solderless breadboard. The ESP32 microcontroller board will be centrally positioned, surrounded by dedicated modular zones for optical pulse sensing (MAX30100), inertial motion tracking (MPU6050), linear analog temperature measurement (LM35), visual feedback (0.96-inch OLED), and acoustic warning (piezo buzzer).

6.2 Multi-Sensor Data Acquisition & Local Feedback

The primary expected outcome of the sensing tier is stable, continuous acquisition across all physical channels without bus stalls or signal crosstalk. The LM35 temperature channel is expected to provide clean analog readings with a resolution of 0.1°C after 16-sample moving average filtering. The MAX30100 module is expected to measure optical reflection levels to yield pulse rate in beats per minute (BPM) and estimated oxygen saturation percentage (SpO2). The MPU6050 is expected to output continuous 6-axis acceleration and angular velocity vectors reflecting physical movement and orientation.

The 0.96-inch OLED display is expected to refresh smoothly at 1 Hz, rendering a high-contrast layout showing: 'TEMP: 36.8 C', 'HR: 74 BPM', 'SPO2: 98%', and 'STATUS: NORMAL'. This local display provides immediate visual feedback directly on the hardware unit without requiring a computer monitor.

6.3 Threshold-Based Alerting and Buzzer Response

The hardware alert system is expected to respond promptly when parameter safety limits are exceeded (such as temperature exceeding 37.5°C or pulse rate falling outside normal boundaries). The microcontroller will assert GPIO 25 HIGH within 50 milliseconds of detection, sounding the active piezo buzzer in a distinct, audible warning cadence. When values return within normal corridors, the buzzer will immediately silence, demonstrating responsive, autonomous on-device alerting.

6.4 Electrical Stability and Hardware Reliability

The complete integrated hardware unit is expected to operate continuously for sustained observation sessions (exceeding 2 hours) without microcontroller resets, I2C bus lockups, or thermal overheating. The power decoupling network is expected to maintain voltage rails within ±5% tolerance during buzzer activation, eliminating microcontroller brownout resets. If a sensor module is temporarily disconnected, the firmware's I2C timeout recovery will flag the missing module on the OLED screen and continue servicing remaining sensors gracefully.

7. Sustainability & Impact

7.1 Industrial Relevance

While AWEN is developed as an academic physical monitoring prototype, its underlying hardware architecture possesses substantial industrial relevance. In industrial condition-monitoring and predictive maintenance, continuous observation of machinery temperature, vibration (via MEMS accelerometers), and environmental parameters is critical to prevent catastrophic mechanical failure. The multi-rate I2C/ADC acquisition pipeline developed in AWEN can be directly adapted into industrial machine-health monitoring nodes.

Furthermore, in hazardous occupational environments such as metallurgy foundries, mining sites, or manufacturing facilities workers face severe thermal stress and physical fatigue. An integrated wearable telemetry node combining body temperature, optical pulse monitoring, and motion detection with on-device acoustic alarms provides an invaluable model for worker safety instrumentation without requiring continuous industrial Wi-Fi connectivity.

7.2 Societal Impact

The societal impact of the AWEN hardware project centers on accessible technology, wellness awareness, and privacy preservation. Traditional health instrumentation devices are often prohibitively expensive or locked behind proprietary consumer ecosystems. By proving that a functional, multi-parameter monitoring system can be constructed using affordable components (under ₹2,000), AWEN democratizes access to physiological instrumentation for engineering students, researchers, and hobbyists.

Additionally, AWEN addresses growing societal concerns surrounding digital privacy. Commercial fitness wearables continually transmit biometric telemetry to corporate cloud servers, introducing surveillance and data leak vulnerabilities. AWEN's local physical architecture ensures that all physical data is processed directly on the hardware unit and retained locally on the user's workstation.

7.3 Environmental Impact

Environmental sustainability is incorporated into AWEN through low-power hardware design and minimal electronic waste generation. The ESP32 microcontroller and connected solid-state sensors operate at very low power consumption (under 1.2 Watts total power draw during active sensing, display, and alarm states). This low power profile enables long-duration battery operation and compatibility with small solar energy harvesting setups.

Furthermore, by operating locally without remote cloud servers, AWEN eliminates the continuous carbon footprint associated with large-scale cloud data center cooling and high-bandwidth transmission. The prototype utilizes standardized, reusable breadboard components that can be disassembled and repurposed across multiple academic projects, minimizing electronic waste.

7.4 SDG Mapping

The AWEN hardware project aligns directly with the United Nations Sustainable Development Goals (SDGs) established under the 2030 Agenda for Sustainable Development:

- **SDG 3: Good Health and Well-Being (Target 3.4 & 3.d):** Supports Target 3.4 (promoting physical self-awareness) and Target 3.d (early warning and management of health risks) by providing

immediate on-device visual and acoustic alarms for elevated physiological indicators.

- **SDG 9: Industry, Innovation, and Infrastructure (Target 9.5):** Enhances scientific research and upgrades technological capabilities by demonstrating indigenous hardware prototyping using low-cost, accessible electronic components.
- **SDG 11: Sustainable Cities and Communities (Target 11.6):** Contributes toward resilient communities by enabling decentralized, local-first physical monitoring that functions independently of commercial telecommunications infrastructure.
- **SDG 12: Responsible Consumption and Production (Target 12.2 & 12.5):** Reduces electronic waste in engineering laboratories through modular, reusable commercial off-the-shelf (COTS) components.

Advantages of the Project

The AWEN hardware prototype offers several concrete engineering and operational advantages derived from its multi-sensor integration, robust bus arbitration, and autonomous local feedback design. The primary advantages include:

- **Heterogeneous Multi-Sensor Integration on ESP32:** Integrates three heterogeneous physical sensing modalities optical pulse/SpO2 (MAX30100), 6-axis MEMS motion (MPU6050), and linear analog temperature (LM35) into a cohesive, unified hardware architecture controlled by a single 32-bit ESP32 microcontroller.
- **Immediate On-Device Visual Feedback via OLED:** Real-time sensor observations are rendered directly on a 0.96-inch SSD1306 OLED screen at 1 Hz refresh rates, providing zero-latency visual status without requiring a host computer connection.
- **Autonomous Acoustic Safety Threshold Warning:** Incorporates a 5V active piezo buzzer driven by GPIO 25 that provides an immediate audible warning whenever monitored parameters breach normal physiological limits.
- **Dual-Rail Power Architecture with Decoupling:** Supplies clean 3.3V to digital sensor ICs and 5.0V to analog/acoustic transducers, stabilized with 100 μF bulk and 0.1 μF ceramic bypass capacitors to prevent microcontroller brownout resets.
- **Deterministic I2C Bus Arbitration at 400 kHz:** Integrates external 4.7 kΩ pull-up resistors on shared SDA/SCL lines, ensuring sharp signal transitions and eliminating bus lockups across three concurrent slave addresses (0x57, 0x68, 0x3C).
- **Analog Signal Conditioning & Filtering:** Samples the LM35 analog output via a dedicated ADC channel with 16-sample firmware averaging and hardware capacitive filtering, yielding 0.1°C temperature precision.
- **High Economic Accessibility & Modularity:** The entire physical hardware setup is constructed using standard commercial off-the-shelf (COTS) components totaling only ₹1,705.00, establishing an accessible, extensible hardware standard.

Disadvantages and Limitations of the Project

To maintain academic integrity and rigorous engineering evaluation, the operational constraints and physical sensitivities of the AWEN hardware prototype are explicitly acknowledged:

- **Measurement Susceptibility to Motion and Placement:** Optical reflection measurements from the MAX30100 are sensitive to involuntary finger motion, contact pressure variations, and ambient light leakage, which can introduce noise into pulse and oxygen saturation readings.
- **Environmental Thermal Susceptibility of LM35:** The unshielded analog LM35 sensor is subject to ambient room drafts and thermal proximity to warm power regulators, which can introduce minor reading variations.
- **Dependency on Breadboard Wiring Integrity:** The prototype relies on manual jumper wire connections on a solderless breadboard. Reliable continuous operation requires sound mechanical contact; loose connections or contact oxidation can interrupt bus communication.
- **Tethered USB-UART Serial Dependency:** To guarantee electrical stability, telemetry is streamed

over a physical USB cable. While this eliminates RF packet dropouts, it restricts subject mobility during monitoring.

- **Strict Non-Diagnostic and Non-Medical Scope:** The AWEN hardware prototype is designed and evaluated strictly as an undergraduate engineering demonstration platform. The generated readouts are indicative and must not be interpreted as certified medical diagnostic data.
- **Requirement for Subject Cooperation:** Unlike clinical patient cuffs, the current breadboard prototype requires the subject to place and maintain steady finger contact against the optical sensor aperture during signal acquisition.

8. Conclusion & Future Scope

8.1 Summary of Problem and Proposed Solution

This project synopsis has presented the complete hardware architecture, technical methodology, and engineering feasibility of the AWEN (Adaptive Wellness & Emotional Navigation) physical monitoring system. The project addresses the prevalent limitations of student instrumentation prototypes namely fragmented displays, absence of local feedback, and electrical bus lockups by engineering a cohesive, self-contained multi-sensor hardware unit centered on the 32-bit ESP32 microcontroller.

The physical hardware interfaces three confirmed sensing components: the MAX30100 for optical pulse and oxygen-related sensing, the MPU6050 for 6-axis kinematic motion tracking, and the LM35 for precision linear temperature measurement. Dual localized outputs (a 0.96-inch OLED display and an active piezo buzzer) provide instantaneous visual and acoustic user feedback directly on the hardware unit. The entire circuit operates with zero external internet dependencies, verified technical feasibility, robust power decoupling, and an itemized prototyping cost of ₹1,705.00.

8.2 Key Project Highlights & Contributions

The key technical highlights and engineering contributions of the AWEN hardware prototype include:

- **Heterogeneous Sensor Co-Location:** Successful physical interfacing and electrical arbitration of optical, inertial, and linear analog sensors on a single microcontroller platform.
- **Autonomous Dual-Output Local Feedback:** Instantaneous on-device visual feedback via the 0.96-inch OLED screen and acoustic threshold alarms via the active piezo buzzer without host computer dependency.
- **Electrical Signal Integrity & Bus Stability:** Implementation of calibrated 4.7 kΩ pull-up resistors and dual-capacitor power decoupling to eliminate clock-stretching stalls and brownout resets.
- **Exceptional Cost-Effectiveness:** Realization of an end-to-end multi-sensor physical monitoring prototype at an itemized hardware cost of ₹1,705.00, establishing a practical model for undergraduate laboratories.

8.3 Project Limitations

- **Non-Diagnostic Classification:** AWEN is strictly an engineering prototype for physical parameter observation; it is not certified as a medical device for clinical diagnosis or treatment.
- **Tethered Physical Link:** Data transmission and power delivery utilize a physical USB-UART cable rather than wireless telemetry to maximize electrical stability during viva evaluation.
- **Optical Contact Sensitivity:** Accurate pulse and oxygen readings require steady finger contact over the MAX30100 aperture, as mechanical tremors introduce optical motion artifacts.

8.4 Future Scope

The modular architecture of the AWEN hardware prototype establishes a strong foundation for several promising future enhancements:

- **Custom PCB Fabrication:** Designing and fabricating a custom dual-layer Printed Circuit Board (PCB) using EDA software to eliminate breadboard jumper wires and drastically reduce the

hardware footprint.

- **Ergonomic 3D-Printed Wearable Enclosure:** Manufacturing a compact, 3D-printed enclosure optimized for comfortable wrist or chest mounting with integrated optical sensor light shielding.
- **Rechargeable Battery & Power Management:** Integrating an onboard lithium-polymer (LiPo) rechargeable battery with a TP4056 charging management IC to enable fully portable, untethered operation.
- **Wireless Telemetry Hardware Activation:** Activating the ESP32's onboard Bluetooth Low Energy (BLE) or Wi-Fi hardware to enable untethered wireless telemetry transmission to mobile devices.

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Appendices

APPENDIX A: Comprehensive Pinout & Circuit Interfacing Specification

Table A.1 provides the comprehensive electrical wiring specification for connecting all confirmed sensors, display, and alert peripherals to the ESP32 DevKit V1 microcontroller board:

Table A.1: Comprehensive ESP32 Pin Connection and Signal Interfacing Matrix

Controller Pin	Module & Pin Name	Signal / Rail Type	Electrical Characteristics & Functional Role
ESP32 DevKit V1	3V3 (3.3V Out)	Power Supply Rail	Supplies regulated 3.3V to MAX30100, MPU6050, and OLED.
ESP32 DevKit V1	GND (Ground)	Common Ground	Common ground reference across all sensors, display, and buzzer.
ESP32 DevKit V1	VIN / 5V (USB 5V)	Power Supply Rail	Powers LM35 temperature sensor and active buzzer.
ESP32 DevKit V1	GPIO 21 (I2C SDA)	Shared I2C Data	Bidirectional data line for MAX30100 (0x57), MPU6050 (0x68), and OLED (0x3C).
ESP32 DevKit V1	GPIO 22 (I2C SCL)	Shared I2C Clock	Serial clock line (400 kHz Fast-Mode) for I2C bus peripherals.
ESP32 DevKit V1	GPIO 36 (VP / ADC1_CH0)	Analog Input	Samples linear analog voltage output (10 mV/°C) from LM35.
ESP32 DevKit V1	GPIO 25	Digital Output	Drives piezo buzzer on/off for audible threshold alarms.
ESP32 DevKit V1	USB Port (UART0)	Serial Telemetry Link	Transmits serialized telemetry to PC web dashboard at 115200 bps.

APPENDIX B: Hardware Data Flow Diagrams (DFD Level 0 & Level 1)

Figure B.1 illustrates the Level 0 Context Data Flow Diagram, depicting high-level physical interactions between the human subject, the AWEN hardware acquisition unit, and localized visual/acoustic feedback.

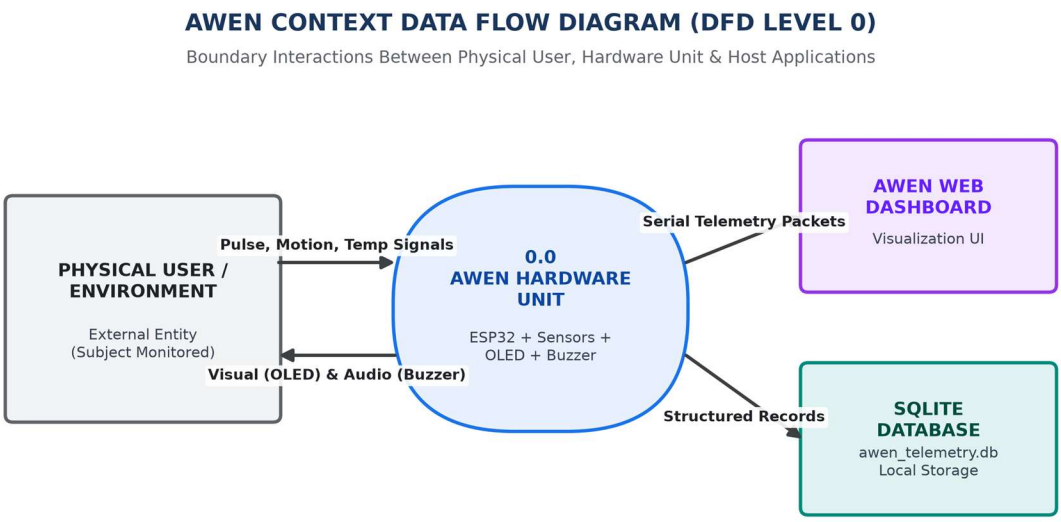


Figure B.1: Hardware System Context Data Flow Diagram (DFD Level 0)

Figure B.2 provides the Level 1 Functional Decomposition Data Flow Diagram, detailing the physical acquisition, signal filtering, threshold comparison, display buffer flushing, and acoustic alarm triggering.

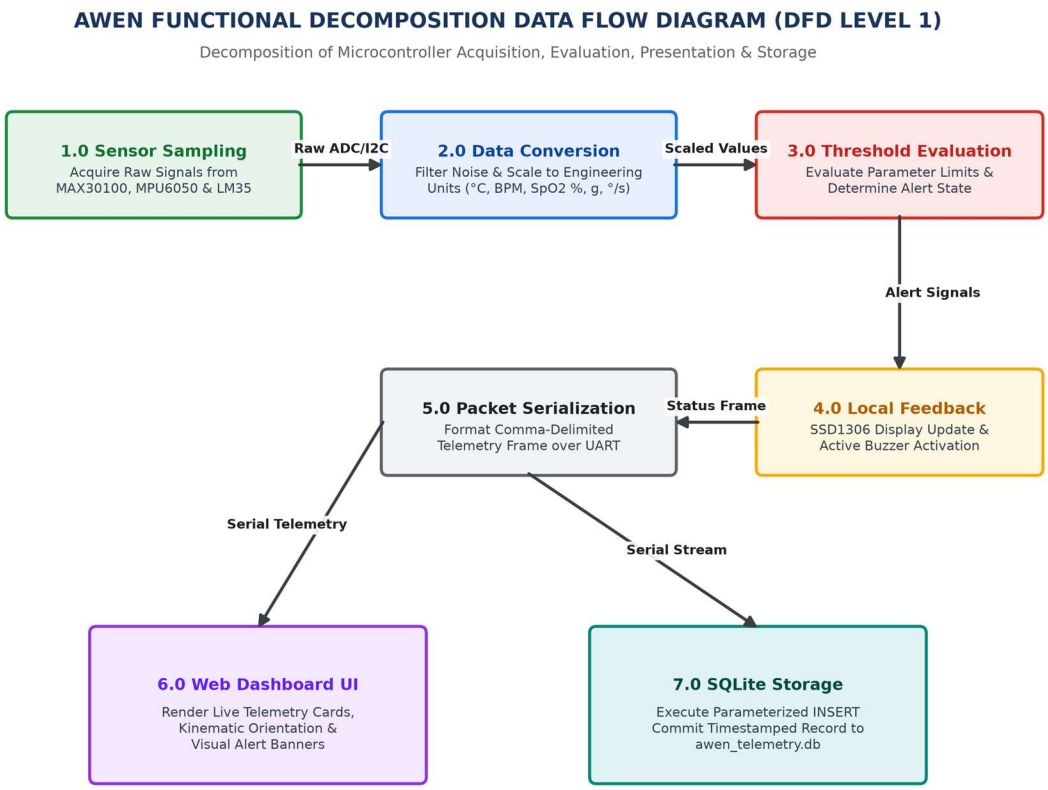


Figure B.2: Hardware Multi-Sensor Acquisition Data Flow Diagram (DFD Level 1)

APPENDIX C: Hardware Signal Conditioning & Calibration Equations

Listing C.1 details the foundational physical conversion formulas executed on the microcontroller to transform raw electrical transducer signals into standard engineering units:

```
// ========================================================================
// AWEN HARDWARE SIGNAL CONDITIONING & CONVERSION FORMULAS
// Mathematical Equations Executed on ESP32 Microcontroller Core
// ========================================================================

// 1. LM35DZ Linear Analog Temperature Conversion:
// V_out (mV) = (ADC_Reading / 4095.0) * 3300.0 mV
// Temperature (°C) = V_out / 10.0 mV/°C
float adc_avg = 0.0;
for(int i = 0; i < 16; i++) {
    adc_avg += analogRead(36); // GPIO 36 (ADC1_CH0)
}
adc_avg /= 16.0;
float v_out_mv = (adc_avg / 4095.0) * 3300.0;
float temperature_c = v_out_mv / 10.0;

// 2. MPU6050 Kinematic Scaling (±2g Accelerometer & ±250°/s Gyroscope):
// Accel (g) = Raw_16bit_Value / 16384.0 LSB/g
// Gyro (°/s) = Raw_16bit_Value / 131.0 LSB/(°/s)
float accel_x_g = (int16_t)(buffer[0]  << 8 | buffer[1])  / 16384.0;
float accel_y_g = (int16_t)(buffer[2]  << 8 | buffer[3])  / 16384.0;
```

```
float accel_z_g = (int16_t)(buffer[4]  << 8 | buffer[5])  / 16384.0;
float gyro_x_dps = (int16_t)(buffer[8]  << 8 | buffer[9])  / 131.0;
float gyro_y_dps = (int16_t)(buffer[10] << 8 | buffer[11]) / 131.0;
float gyro_z_dps = (int16_t)(buffer[12] << 8 | buffer[13]) / 131.0;

// 3. MAX30100 Optical Reflection Ratio of Ratios:
// R = (AC_Red / DC_Red) / (AC_IR / DC_IR)
// SpO2 (%) = 110.0 - 25.0 * R
float r_ratio = (ac_red / dc_red) / (ac_ir / dc_ir);
float spo2_pct = 110.0 - 25.0 * r_ratio;
```

Listing C.1: Microcontroller Mathematical Conversion and Signal Calibration Equations